

6-Day Intensive Training Plan

Django + React + MongoDB + Docker + AWS

Project Theme: Task Management System

You will build a full-stack application using Django (Backend), MongoDB (Database), React (Frontend), Docker (Containerization), Git (Version Control), and AWS (Deployment). This 6-day intensive program is hands-on and output-driven.

DAY 1 – Environment Setup + Git + MongoDB + React Setup

- You will install and configure MongoDB locally.
- You will create database 'intern_db' and 'users' collection.
- You will create index on email field.
- You will learn basic Linux shell commands and create cron job.
- You will setup Git repository, create branch, raise PR and resolve conflict.
- You will setup React project and create basic components.

DAY 2 – Django Backend + API + React Integration

- You will create Django project and connect to MongoDB.
- You will build CRUD REST APIs.
- You will test APIs using Postman.
- You will integrate React frontend with backend using Axios.

DAY 3 – Best Practices + Security

- You will implement compound indexing and query optimization.
- You will take MongoDB backup using mongodump and restore using mongorestore.
- You will implement exception handling and JWT authentication in Django.
- You will structure scalable React project with custom hooks.

DAY 4 – Docker + Unit Testing

- You will write Dockerfile for Django and React.
- You will configure docker-compose with MongoDB container.
- You will write Django unit tests for models and APIs.

- You will write React unit tests using Jest.

DAY 5 – AWS Deployment

- You will launch EC2 instance and configure security groups.
- You will deploy backend on EC2.
- You will host frontend on S3.
- You will push Docker image to ECR.
- You will configure IAM roles and monitor using CloudWatch.

DAY 6 – CI/CD + Final Evaluation

- You will configure GitHub Actions for CI pipeline.
- You will automate testing and Docker builds on PR.
- You will configure logging and monitoring.
- You will submit final project with documentation and deployment.

Evaluation Criteria

Criteria	Marks
Code Quality	20
API Design	15
Database Design	15
Testing	15
Dockerization	15
AWS Deployment	10
Documentation	10